

VIJETA SINGH YADAV

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EDUCATION

MNNIT Allahabad • July 2019 – July 2023

B.Tech in Information Technology • GPA: 8.1/10

WORK EXPERIENCE

Oracle Software Developer 2

Sep 2024 – Present

- Contributed to **RevAI**, enabling natural language querying over enterprise datasets using LLM pipelines. Designed validation and execution layers to ensure safe query generation and prevent unsafe DB operations
- Built a scalable search and pagination service enabling infinite scroll over large datasets, improving data retrieval efficiency by **37%**
- Designed and implemented a secure **LLM to SQL** pipeline with validation and execution guardrails, ensuring safe database interactions in production systems
- Built a semantic data layer with feedback-driven optimization, improving query accuracy and enabling self-serve analytics
- Engineered high-performance data ingestion for tabular inputs (Excel/CSV/SQL) with dynamic schema mapping and real-time validation

Posha SDE 1

Mar 2024 – Sep 2024

- Implemented an asynchronous, multi-threaded upload framework with idempotency checks and exponential backoff retries, ensuring reliable and consistent log delivery
- Built a fault-tolerant real-time logging pipeline with segmentation (time/size/event-based), local persistence, and resumable uploads, ensuring **zero data loss and reliable recovery** from partial writes, corruption, and system failures
- Designed state-driven execution logic for recipe workflows, enabling safe instruction transitions using session state, iteration tracking, and strict validation of system states (e.g., paused vs running)
- Ensured reliable and observable behavior by handling invalid state transitions, integrating structured logging, and validating flows through high test coverage (**>85%**)
- Built a personalized pod arrangement optimization system using historical session data (via GraphQL), improving user efficiency and reducing redundant setup steps

MathWorks – Associate Software Engineer

Jan 2023 – Mar 2024

- Built a Java-based plugin (Stapler + Jenkins) reducing MATLAB project build/test setup time by **90%**
- Implemented MQTT-based communication with support for Last Will and Testament (LWT) to enable automatic failure detection and improve system reliability in multi-device environments and increased throughput by **30%**
- Developed a recommendation system achieving **93%** accuracy for context-free suggestions in Simulink
- Optimized large-scale data processing using TF-IDF, reducing trivial transactions by **82%**

MathWorks – Software Engineer Intern

June 2022 – July 2022

- Refactored search and indexing pipeline for Simulink Model Finder, reducing memory usage by **13%**
- Improved ranking heuristics using design patterns, enhancing relevance and maintainability

PROJECTS

RAG-Based Semantic Search & Real-Time Recommendation Engine

GitHub

- Architecture & Real-time Processing:** Built platform using Spring Boot with polyglot persistence (PostgreSQL, MongoDB, Elasticsearch, Redis) and event-driven architecture with Apache Kafka
- AI-Powered Semantic Search:** Developed RAG system using Ollama LLMs with vector embeddings and pgvector, achieving 85%+ relevance and <100ms query latency while processing 1000+ events/minute
- Performance Optimization:** Achieved <200ms API response times via caching, async processing, and hybrid search combining Elasticsearch with semantic vector search and Kibana analytics

TECHNICAL SKILLS

Languages: Java, Python, C++, Javascript
Backend: Spring Boot, REST APIs, GraphQL
Data & Infra: Kafka, Redis, MongoDB, Elasticsearch
AI Systems: RAG, LLM pipelines, Vector DB (pgvector)
Core: System Design, Concurrency, Distributed Systems